



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2018 – 0378



Agricultural Machinery – Power-Operated Corn Sheller ACT SUPER

Test Report Number	2018 – 0378
Date of Issuance	July 1, 2019
Valid Until	July 1, 2024
Agricultural Machinery	Power-operated corn sheller
Type	Axial-flow with sifter and fan, open cylinder
Brand	ACT
Model	SUPER
Test Requested by	ACT MACHINERIES & METALCRAFT CORPORATION Turayong, Cauayan City, Isabela

SUMMARY

Performance Criteria	Requirement as per PAES 208:2000	AMTEC Test
Shelling Recovery, %, minimum	97.0	99.75
Shelling Efficiency, %, minimum	99.5	100
Losses, %, maximum		
a) Blower Loss	1.0	0.09
b) Separation Loss	1.0	0.16
c) Unshelled Loss	0.5	0
d) Scattering Loss	0.5	0
Purity, %, minimum	98	99.1
Mechanically Damaged Kernel, %, maximum	3.0	2.84
Net Cracked Kernel, %, maximum	5.0	4.33
Noise Level, dB(A), maximum	100*	99.6

Note: *Allowable noise level for six (6) hours of continuous exposure based on Occupational Safety and Health Standards, Ministry of Labor, Philippines, 1983.

OBSERVATIONS

1. Operator's manual	None
2. Previous valid test reports of similar brand and model	None
3. Prime mover used	9.19 kW GOLDEN BOW 125 D1 Diesel Engine
4. Manufacturer	ACT MACHINERIES & METALCRAFT CORPORATION
5. Serial number	6994-18
6. Transporting of machine	Provided with transport wheels, hitch point and towing bars.
7. Adjustment of parts	Engine cylinder belt tension can be adjusted. Air flow can be adjusted.
8. Safety features	No cover was provided in the V-belt-pulley transmission system.
9. Cleaning of concave and cylinder	With cover hinged to the body cleaning is possible.
10. Cleaning of fan component	Fan casing is fixed which make cleaning difficult.
11. Failure or abnormalities that observed on the sheller or its components during and after shelling operation	None
12. Number of operators	Six (two for hauling of test materials to the sheller, one for feeding of test materials, two for collecting and bagging of shelled grains, and one for stirring of mixture of grains and cobs on the oscillating screen)
13. Other remarks	No breakdown occurred.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date/s of Test	October 10, 2019
Location of Test	Turayong, Cauayan City, Isabela
Items Tested	Shelling capacity, Shelling recovery, Shelling efficiency, Quality of work, Noise level, Fuel consumption, Speed of rotating components, and Number of operators

METHODS OF TEST

The corn husker/sheller was tested based on the Philippine National Standards PNS/PAES 209:2000 Agricultural Machinery – Power-Operated Corn Sheller – Methods of Test.

DESCRIPTION OF THE MACHINE



Figure 1. Parts of ACT SUPER Power-Operated Corn Sheller powered by 9.19 kW GOLDEN BOW 125 D1 Diesel Engine:
(a) Feeding table, (b) Prime mover, (c) Belt-pulley transmission system, and (d) Oscillating screen

SPECIFICATIONS

Item		Manufacturer's Specification*	AMTEC Verification
1	Dimension and weight of sheller		
1.2	Overall length, mm	ND	1910
1.3	Overall width, mm	ND	1900
1.4	Overall height, mm	ND	1945
1.5	Weight of machine without engine, kg	ND	NM
2	Crop(s) for which machine is suitable	ND	Corn
3	Rated output capacity, kg/h	ND	4651
4	Recommended cylinder speed, rpm	ND	782
5	Engine		
5.1	Brand	ND	GOLDEN BOW
5.2	Model	ND	125DI
5.3	Serial number	ND	None
5.4	Make	ND	Thailand
5.5	Rated power, kW	ND	9.19
5.6	Rated speed, rpm	ND	2400
5.7	Type	ND	Four-stroke, horizontal single cylinder diesel engine
5.8	Weight	ND	NM
5.9	Starting system	ND	Manual cranking lever
5.10	Cooling system	ND	Water-cooled
6	Type of power transmission system		
6.1	Engine to Shelling cylinder	V-Belt & Pulley	
6.1.1	Engine pulley diameter	ND	102 x 2B
6.1.2	Shelling cylinder pulley diameter	ND	254 x 2B
6.1.3	Belt size	ND	B-66

Note: *The requesting party did not supply the complete machine specifications

ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification*	AMTEC Verification
6.2	Shelling cylinder to oscillating screen and fan	V-Belt & Pulley	
6.2.1	Shelling cylinder pulley diameter	ND	127 x 1B
6.2.2	Oscillating screen pulley diameter	ND	179 x 1B
6.2.3	Fan pulley diameter	ND	76 x 1B
6.2.4	Belt size	ND	B-52
6.3	Fan to auger	V-Belt & Pulley	
6.3.1	Fan pulley diameter	ND	76 x 1B
6.3.2	Auger pulley diameter	ND	179 x 1B
6.3.3	Belt size	ND	ND
7	Shelling cylinder		
7.1	Type	ND	Open cylinder
7.2	Size, L x D, mm	ND	990 x 475
7.3	Cylinder teeth		
7.3.1	Type	ND	Peg-teeth (knife bar)
7.3.2	Size, L x W x T, mm	ND	55 x 25 x 9
7.3.3	Number/Row	ND	8, 8, and 9
7.3.4	No. of rows	ND	3
7.3.5	Arrangement	ND	Alternate
7.3.6	Means of attachment	ND	Welded
7.3.7	Material	ND	Mild steel
8	Fan		
8.1	Type	ND	Radial centrifugal
8.2	No. of units	ND	2
8.3	Material	ND	G.I. Sheet, Gage 16
8.3	Impeller		
8.3.1	Number of blades	ND	4
8.3.2	Length, mm	ND	410

Note: *The requesting party did not supply the complete machine specifications

ND – No Data

SPECIFICATIONS (Cont'n)

Item	Manufacturer's Specification*	AMTEC Verification
9 Oscillating screen		
9.1 Shape	ND	Rectangular
9.1 Dimension, L x W, mm	ND	1200 x 610
9.2 Size of perforations, mm	ND	12
9.3 Length of stroke, mm	ND	15
9.4 Material	ND	Perforated G.I. Metal Sheet
10 Concave component		
10.1 Spacing between grills, mm	ND	20
10.2 Clearance between concave and peg teeth, mm	ND	20
10.3 Material	ND	9 mm round bar
11 Feeding table		
11.1 Shape	ND	Rectangular
11.2 Dimensions, L x W, mm	ND	1200 x 610
11.3 Height from the ground, mm	ND	1250
11.5 Mode of attachment	ND	Bolted
11.6 Material	ND	G.I Sheet, gage 18
12 Transport device		
12.1 Type	ND	Pneumatic tires
12.2 Size	ND	NM
13 Safety device	ND	None
14 Discharge device	ND	Auger Mechanism
15 Number of operator	ND	6
16 Adjustment	ND	Engine cylinder belt tension can be adjusted. Air flow can be adjusted.
17 Other special features	ND	Provide with hitch point and towing bars.

Note: *The requesting party did not supply the complete machine specifications

ND – No Data, NM – Not Measured

RESULTS

Item		Data	
1	Crop condition		
1.1	Kind/Variety	Unknown	
1.2	Days after harvest	-	
1.3	Moisture content, %wb	23.13	
1.4	Size of corn ear, L x D, mm	246 x 55	
1.5	Kernel-corn ear ratio	0.79	
2	Performance test		
2.1	Speed of components, rpm	Without load	With load
2.2	Prime mover	2615	2146
2.3	Shelling cylinder shaft	1021	782
2.4	Cylinder peripheral speed, m/min	1522	1166
2.5	Fan shaft	1982	1610
2.	Oscillating screen shaft	859	669
3	Fan air velocity, suction (without load), m/s	3.71	
4	Noise level, dB(A)		
4.1	Feeding operator	89.0	99.6
4.2	Bagger	87.6	99.1
5	Shelling time, min	9	
6	Shelled kernel, kg	698	
7	Shelling capacity, kg/h		
7.1	Actual	4651	
7.2	Corrected	4429	
8	Fuel time, min	19.4	
9	Fuel consumed, mL	920	
10	Fuel consumption rate, L/h	2.88	

RESULTS (Cont'n)

LABORATORY ANALYSIS

Item		Data
1	Shelling Recovery, %	99.75
2	Shelling Efficiency, %	100
3	Blower Loss, %	0.09
4	Unshelled Loss, %	0
5	Separation Loss, %	0.16
6	Scattering Loss, %	0
7	Purity of Shelled Kernel, %	99.1
8	Net Cracked Kernel, %	4.33
8.1	Manually Shelled, %	0
8.2	Machine Shelled	4.33
9	Mechanically Damaged Kernel, %	2.84

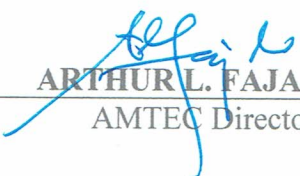
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